O-RAN.WG3.Y1AP-R004-v01.01

Technical Specification

O-RAN Work Group 3 (Near-Real-time RAN Intelligent Controller and E2 Interface)

Y1 interface: Application Protocol

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Contents

Foreword

This Technical Specification (TS) has been produced by WG3 of the O-RAN Alliance.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.

yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.

zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the stage3 definition of the application protocols of the Y1 services.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE 1: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

NOTE 2: In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in 3GPP Release 17.

The following referenced documents are necessary for the application of the present document.

[1] O-RAN.WG3.Y1GAP, "Y1 interface: General Aspects and Principles".

[2] O-RAN.WG3.Y1TD, "Y1 interface: Type Definitions".

[3] OpenAPI, "OpenAPI 3.1.0 Specification", https://github.com/OAI/OpenAPI-Specification/blob/master/versions/3.1.0.md.

[4] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

[5] IETF RFC 7807: "Problem Details for HTTP APIs".

[6] IETF RFC 3986: "Uniform Resource Identifier (URI): Generic Syntax".

[7] 3GPP TS 29.122: "T8 reference point for Northbound APIs".

[8] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces; Stage 3".

[9] O-RAN.WG1.OAD, "O-RAN Architecture Description".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE 1: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

NOTE 2: In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in 3GPP Release 17.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

[i.1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications"

[i.2] Semantic Versioning Specification. Available at https://semver.org.

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [i.1] and O-RAN.WG1.OAD [9] apply.=

3.2 Symbols

For the purposes of the present document, the symbols given in 3GPP TR 21.905 [i.1] and O-RAN.WG1.OAD [9] apply.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [i.1], O-RAN.WG1.OAD [9] and the following apply:

AS	Application Server
RAI	RAN Analytics Information
SCEF	Service Capability Exposure Function
SCS	Services Capability Server

4 General

4.1 Introduction

The present document contains the stage-3 definitions for the services identified in Y1GAP [1]. Certain data types, including but not limited to RAI data types are defined in Y1TD [2].

One or more solution sets in terms of network protocol and data interchange format may be defined for each Y1 service API. The use of a specific solution set for a Y1 service API may be preconfigured or selected via negotiation mechanisms between the Near-RT RIC and the Y1 consumer.

4.2 Versioning of Y1 service APIs

4.2.1 General rules for API version

Each Y1 service API is versioned independently. An API version shall consist of at least 3 fields, following a MAJOR.MINOR.PATCH pattern according to the Semantic Versioning Specification [i.2] unless specified otherwise.

EXAMPLE: "1.0.0".

The basic rules as reproduced from [i.2] shall apply for incrementing the versions:

- MAJOR version is incremented if backward incompatible change is made.
- MINOR version is incremented if functional change is made in a backward compatible way.
- PATCH version is incremented if correction is made in a backward compatible way.

4.2.2 Versioning for multiple solution sets of a Y1 service API

For a single Y1 service API in the present document, a single version should be kept across all the solution sets defined for that API. That means the different solution sets for a Y1 service API should evolve simultaneously.

4.2.3 Versioning for certain data types

The Y1 service APIs are designed in the approach that certain data types are decoupled from the APIs, and such data types may have versions independent from the API version. To that end, the full compatibility for a given solution set of a single Y1 service API between Near-RT RIC and Y1 consumer is jointly governed by the API version and the relevant data type version(s).

5 Usage of solution sets

5.1 Introduction

For each solution set, the generic designs are defined in this clause.

The solution sets that may be used for a Y1 service API are listed in Table 5.1-1.

Table 5.1-1: Solution sets for Y1 interface

Solution Set	Clause
Solution 1: HTTP REST with JSON	5.2

5.2 Usage of HTTP REST with JSON

5.2.1 General

The solution set of HTTP REST with JSON is introduced in clause 7.2 of Y1GAP [1].

The OpenAPI [3] specifications for each Y1 service API with this solution set are documented in Annex A.

5.2.2 Content type

The bodies of HTTP request and successful HTTP responses shall be encoded in JSON format. The MIME media type that shall be used within the related Content-Type header field is "application/json", as defined in IETF RFC 8259 [4], unless specified otherwise.

The "Problem Details" JSON object used to indicate additional details of the error in a HTTP response body shall be signaled by the content type "application/problem+json", as defined in IETF RFC 7807 [5].

5.2.3 URI structure

5.2.3.1 Resource URI structure

The resource URIs of all Y1 service APIs, unless specified otherwise, shall be structured as:

{apiRoot}/<apiName>/<apiMajorVersion>/<apiSpecificResourceUriPart>

with the following components:

- The {apiRoot} indicates the scheme, the host name and optional port, and an optional implementation-specific sequence of path segments (starting with "/").
- The <apiName> indicates the API name of the service API in an abbreviated form. It is defined in the clause specifying the corresponding Y1 service API.

- The <apiMajorVersion> indicates the major version of the service API (see clause 4.2.1) and is defined in the clause specifying the corresponding Y1 service API.
- Each <apiSpecificResourceUriPart> represents a specific resource of the service API. It is defined in the clause specifying the corresponding Y1 service API.

All resource URIs of the service APIs shall comply with the URI syntax as defined in IETF RFC 3986 [6].

5.2.4 HTTP headers

5.2.4.1 General

The HTTP headers used in [7] shall be supported by all Y1 service APIs with the solution set "HTTP REST with JSON" in the present document, unless specified otherwise.

5.2.5 Error handling

5.2.5.1 General

Unless specified otherwise, the HTTP error handling mechanism described in clause 5.2.6 of [7] shall be supported by all Y1 service APIs with the solution set "HTTP REST with JSON" in the present document, with the following replacements:

- the SCEF is replaced by the Near-RT RIC; and
- the SCS/AS is replaced by the Y1 consumer.

6 API Definitions

6.1 Introduction

The Y1 service APIs defined in the present document are listed in Table 6.1-1.

Table 6.1-1: List of Y1 Service APIs

Y1 service API	Clause	API version	Supported solution sets
Y1_RAI_Subscription	6.2	1.0.0	HTTP REST with JSON
Y1_RAI_Query	6.3	1.0.0	HTTP REST with JSON

6.2 Y1_RAI_Subscription API

6.2.1 Introduction

The Y1_RAI_ Subscription API is used to access the Y1_RAI_Subscription service.

The service operations and corresponding API definitions are organized in the following clauses for different solution sets.

6.2.2 HTTP REST with JSON solution set

6.2.2.1 Service operations

6.2.2.1.1 Introduction

Table 6.2.2.1.1-1 lists the operations supported by the Y1_RAI_Subscription service for the solution set "HTTP REST with JSON".

Table 6.2.2.2.1-1: Operations of the Y1_RAI_Subscription service

Service operation name	Description	Initiated by
RAI_Subscribe	This service operation is used by a Y1 consumer to subscribe or update subscription for RAI notifications. Periodic or event-triggered notification can be subscribed.	Y1 consumer
RAI_Unsubscribe	This service operation is used by a Y1 consumer to unsubscribe from RAI notifications.	Y1 consumer
RAI_Notify	This service operation is used by a Near-RT RIC to notify Y1 consumes about subscribed RAI.	Near-RT RIC

6.2.2.1.2 RAI_Subscribe service operation

6.2.2.1.2.1 General

The RAI_Subscribe service operation is used by a Y1 consumer to create or update a subscription for RAI notifications from the Near-RT RIC.

6.2.2.1.2.2 Subscribe to RAI notifications

Figure 6.2.2.1.2.2-1 shows a scenario where the Y1 consumer sends a request to the Near-RT RIC to create a subscription to RAI notifications. The operation is based on HTTP POST.

```
@startuml
participant "Y1 consumer" as cons
participant "Near-RT RIC" as prod
cons ->> prod: 1. POST .../subscriptions
prod ->> cons: 2. 201 Created
@enduml
```

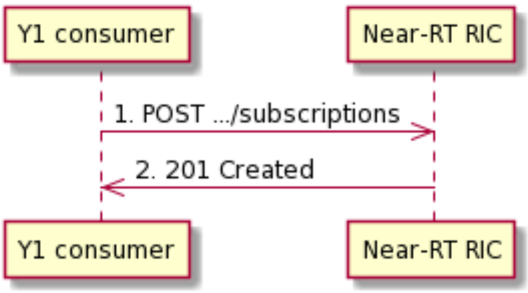

Figure 6.2.2.1.2.2-1: Y1 consumer subscribes to notifications

The Y1 consumer shall send an HTTP POST request to the Near-RT RIC with the Resource URL "{apiRoot}/Y1_RAI_Subscription/<apiMajorVersion>/subscriptions" representing the "All Y1 RAI Subscriptions" resource, to create an "Individual Y1 RAI Subscription" resource according to the information in the message.

NOTE: The path segment " Y1_RAI_Subscription" does not follow the related naming convention specified in clause 5.2.4.1 of [7]. The path segment is however kept as currently defined in this specification for backward compatibility considerations.

The "RaiSubscription" data structure provided in the request body shall include:

- the type of RAI as the "raiType" attribute;
- the version of the RAI type as the "raiTypeVersion" attribute;
- the target entity/entities of the RAI in the "targetEntities" attribute;
- an URI where to receive the RAI notifications as the "notificationTargetAddress" attribute;
- the timing or the events when the RAI notifications are triggered in the "notificationCriteria" attribute, which shall include:
  - the method used to trigger RAI notifications as the "notificationMethod" attribute;
  - the time interval between periodic RAI notifications as the "notificationPeriod" attribute, if the "notificationMethod" attribute is set to "PERIODIC";
  - the trigger event that triggers RAI notifications in the "notificationTriggerEvent" attribute, if the "notificationMethod" attribute is set to "EVENT_TRIGGERED";

  and may include:

  - the expected time of the first RAI notification as the "notificationStartTime" attribute, if the "notificationMethod" attribute is set to "PERIODIC".

The "RaiSubscription" data structure provided in the request body may include:

- the expected validity period(s) for the RAI in the "expectedValidityPeriodsRelative" attribute; an expected validity period before the occurrence of a RAI notification refers to statistics for a past time, and an expected validity period after the occurrence of a RAI notification refers to a prediction for a future time.
- the filter parameters to filter RAI contents in the RAI report as the "filterParameters" attribute;

Upon the reception of the HTTP POST request, the Near-RT RIC shall create the subscription according to the request, assign a unique subscription ID as "subscriptionId", and store the record.

If the Near-RT RIC successfully creates an "Individual Y1 RAI Subscription" resource, the Near-RT RIC shall respond with "201 Created" status code, with the message body containing a representation of the created subscription. The response shall include a "Location" HTTP header field which contains the URI of the created subscription, i.e., "{apiRoot}/Y1_RAI_Subscription/<apiMajorVersion>/subscriptions/{subscriptionId}".

If an error occurs when processing the HTTP POST request, the Near-RT RIC shall send an HTTP error response as specified in clause 6.2.2.6. In particular,

- If the necessary data to perform the analysis are unavailable, the Near-RT RIC shall reject the request with a "500 Internal Server Error" status code and a "ProblemDetails" data structure including the "cause" attribute set to "DATA_UNAVAILABLE".

6.2.2.1.2.3 Update subscription for RAI notifications

Figure 6.2.2.1.2.3-1 shows a scenario where the Y1 consumer sends a request to the Near-RT RIC to update a subscription to RAI notifications. The operation is based on HTTP PUT.

```
@startuml
participant "Y1 consumer" as cons
participant "Near-RT RIC" as prod
cons ->> prod: 1. PUT .../subscriptions/{subscriptionId}
prod ->> cons: 2a. "200 OK" or 2b. "204 No Content"
```



```
@enduml
```

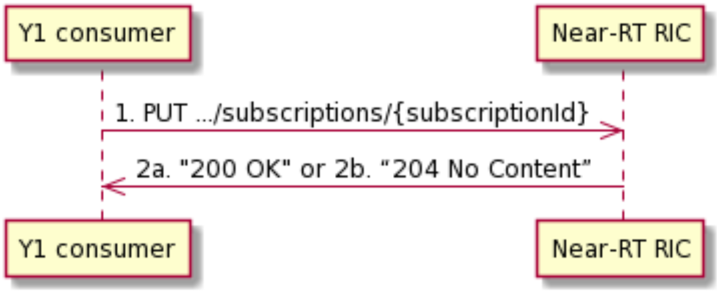

Figure 6.2.2.1.2.3-1: Y1 consumer updates subscription to notifications

The Y1 consumer shall send an HTTP PUT request to the Near-RT RIC with the Resource URL "{apiRoot}/Y1_RAI_Subscription/<apiMajorVersion>/subscriptions/{subscriptionId}" representing the "Individual Y1 RAI Subscription" resource, to update that resource identified by the "subscriptionId" field.

The "RaiSubscription" data structure provided in the request body shall include the same contents as described in clause 6.2.2.1.2.2.

Upon the reception of the HTTP PUT request, the Near-RT RIC shall update the subscription according to the request, and store the record.

If the Near-RT RIC successfully updates the "Individual Y1 RAI Subscription" resource, the Near-RT RIC shall respond with:

- a "200 OK" status code with the message body containing a representation of the updated subscription; or,
- a "204 No Content" status code.

If an error occurs when processing the HTTP PUT request, the Near-RT RIC shall send an HTTP error response as specified in clause 6.2.2.6. In particular,

- If the necessary data to perform the analysis are unavailable, the Near-RT RIC shall reject the request with a "500 Internal Server Error" status code and a "ProblemDetails" data structure including the "cause" attribute set to "DATA_UNAVAILABLE".

6.2.2.1.3 RAI_Unsubscribe service operation

6.2.2.1.3.1 General

The RAI_Unsubscribe service operation is used by an Y1 consumer to delete a subscription for RAI notifications.

6.2.2.1.3.2 Unsubscribe from RAI notifications

Figure 6.2.2.1.3.2-1 shows a scenario where the Y1 consumer sends a request to the Near-RT RIC to unsubscribe from RAI notifications. The operation is based on HTTP DELETE.

```
@startuml
participant "Y1 consumer" as cons
participant "Near-RT RIC" as prod
cons ->> prod: 1. DELETE .../subscriptions/{subscriptionId}
prod ->> cons: 2. 204 No Content
@enduml
```


Figure 6.2.2.1.3.2-1: Y1 consumer unsubscribes from notifications

The Y1 consumer shall send an HTTP DELETE request to the Near-RT RIC with the resource URI "{apiRoot}/ Y1_RAI_Subscription/<apiMajorVersion>/subscriptions/{subscriptionId}" representing the "Individual Y1 RAI Subscription" resource, to delete that resource identified by the "subscriptionId" field.

The message body shall be empty.

Upon the reception of the HTTP DELETE request, the Near-RT RIC shall remove the "Individual Y1 RAI Subscription" resource according to the request.

If the Near-RT RIC successfully removes the "Individual Y1 RAI Subscription" resource, the Near-RT RIC shall respond with "204 No Content" status code.

If an error occurs when processing the HTTP DELETE request, the Near-RT RIC shall send an HTTP error response as specified in clause 6.2.2.6.

6.2.2.1.4 RAI_Notify service operation

6.2.2.1.4.1 General

The RAI_Notify service operation is used by the Near-RT RIC to send RAI notifications to the Y1 consumer, or to terminate the subscription.

6.2.2.1.4.2 Notify subscribed RAI or termination of subscription

Figure 6.2.2.1.4.2-1 shows a scenario where the Near-RT RIC sends a request to the Y1 Consumer to send RAI notifications or to terminate the subscription. The operation is based on HTTP POST.

```
@startuml
participant "Y1 consumer" as prod
participant "Near-RT RIC" as cons
cons ->> prod: 1. POST{notificationTargetAddress}
prod ->> cons: 2. "204 No Content"
@enduml
```

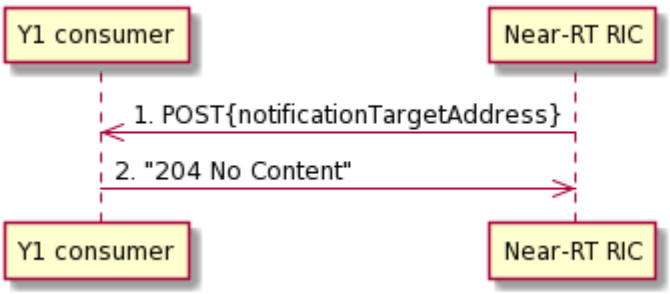

Figure 6.2.2.1.4.2-1: Y1 consumer notifies the subscribed event

The Near-RT RIC shall send an HTTP POST request to the Y1 consumer with "{notificationTargetAddress}" as the resource URL.

The request shall be triggered according to the notification criteria as the "notificationCriteria" attribute in the subscription request, or when Near-RT RIC determines to terminate the subscription.

When the request is triggered by the notification criteria, the "RaiNotification" data structure provided in the request body shall include:

- the subscription ID as the "subscriptionId" attribute;
- the RAI reports in the "raiReports" attribute that, for each RAI report, the "RaiReport" data type shall include:
  - the target entity of the RAI report as the "targetEntity" attribute;
  - the validity period of the RAI report as the "validityPeriod" attribute;
  - the actual RAI contents as the "raiContents" attribute;

  and may include:

  - the confidence of the RAI report, if the RAI report is a prediction;
  - the timestamp when Near-RT RIC generates the RAI report as the "timeStampRaiGeneration" attribute.

When the request is to terminate the subscription, the "RaiNotification" data structure provided in the request body shall include:

- the subscription ID as the "subscriptionId" attribute;
- the termination indication as the "raiSubscriptionTerminationIndication" which is set to "TRUE";

and may include:

- the reason for which Near-RT RIC terminates the subscription.

Upon the reception of the HTTP POST request, the Y1 consumer shall process the information in the message.

If the Y1 consumer successfully processes the request, the Y1 consumer shall respond with "204 No Content" status code.

If errors occur when processing the HTTP DELETE request, the Y1 consumer shall send an HTTP error response as specified in clause 6.2.2.6.

6.2.2.2 Resources

6.2.2.2.1 Resource URI structure

The resource URI shall have the structure as defined in clause 5.2.3.1 with the following clarifications:

- The <apiName> shall be "Y1_RAI_Subscription".
- The <apiMajorVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as illustrated in figure 6.2.2.2.1-1.

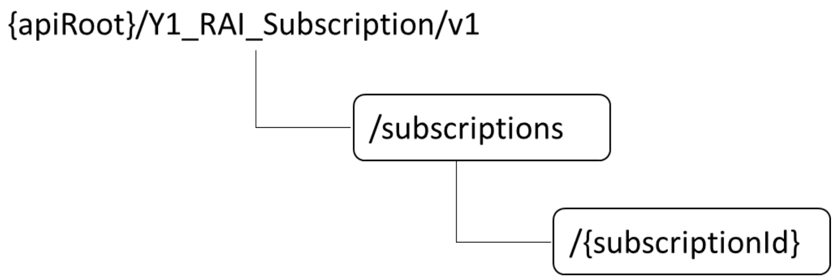

Figure 6.2.2.2.1-1: Resource URI structure of the Y1_RAI_Subscription API

Table 6.2.2.2.1-1 provides an overview of the resources and applicable HTTP methods for the Y1_RAI_Subscription API.

Table 6.2.2.2.1-1: Resources and methods overview

Resource name	API specific resource URI part	HTTP method or custom operation	Description
All Y1 RAI Subscriptions	/subscriptions	POST	Creates a new Individual Y1 RAI Subscription resource.
Individual Y1 RAI Subscription	/subscriptions/{subscriptionId}	DELETE	Deletes an Individual Y1 RAI Subscription resource identified by {subscriptionId}.
		PUT	Updates an existing Individual Y1 RAI Subscription resource.

6.2.2.2.2 Resource: All Y1 RAI Subscriptions

6.2.2.2.2.1 Description

The All Y1 RAI Subscriptions resource represents all subscriptions to the Y1_RAI_Subscription service at a given Near-RT RIC. The resource allows a Y1 consumer to create a new Individual Y1 RAI Subscription resource.

6.2.2.2.2.2 Resource Definition

Resource URI: **{apiRoot}/Y1_RAI_Subscription/v1/subscriptions**

This resource shall support the resource URI variables defined in table 6.2.2.2.2.2.

Table 6.2.2.2.2.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 5.2.3.1

6.2.2.2.2.3 Resource Standard Methods

6.2.2.2.2.3.1 HTTP POST

This method shall support the URI query parameters specified in table 6.2.2.2.2.3.1-1.

Table 6.2.2.2.2.3.1-1: URI query parameters supported by the POST method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.2.2.2.3.1-2 and the response data structures and response codes specified in table 6.2.2.2.2.3.1-3.

Table 6.2.2.2.2.3.1-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
RaiSubscription	M	1	Create a new Individual Y1 RAI Subscription resource.

Table 6.2.2.2.2.3.1-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
RaiSubscription	M	1	201 Created	The creation of an Individual Y1 RAI Subscription resource is confirmed and a representation of that resource is returned.
ProblemDetails	O	0..1	500 Internal Server Error	(NOTE 2)
NOTE 1: The mandatory HTTP error status codes for the POST method used by [7] also apply. See clause 5.2.5. NOTE 2: Failure cases are described in clause 6.2.2.6.				

Table 6.2.2.2.2.3.1-4: Headers supported by the 201 Response Code on this resource

Name	Data type	P	Cardinality	Description
Location	string	M	1	Contains the URI of the newly created resource, according to the structure: {apiRoot}/Y1_RAI_Subscription/v1/subscriptions/{subscriptionId}

6.2.2.2.4 Resource Custom Operations

No custom operations are defined in the present document.

6.2.2.2.3 Resource: Individual Y1 RAI Subscription

6.2.2.2.3.1 Description

The Individual Y1 RAI Subscription resource represents a single subscription to the Y1_RAI_Subscription service via Y1 interface.

6.2.2.2.3.2 Resource definition

Resource URI: **{apiRoot}/Y1_RAI_Subscription/v1/subscriptions/{subscriptionId}**

This resource shall support the resource URI variables defined in table 6.2.2.2.3.2-1.

Table 6.2.2.2.3.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 5.2.3.1
subscriptionId	string	Identifies a subscription to the Y1_RAI_Subscription service

6.2.2.2.3.3 Resource Standard Methods

6.2.2.2.3.3.1 HTTP DELETE

This method shall support the URI query parameters specified in table 6.2.2.2.3.3.1-1.

Table 6.2.2.2.3.3.1-1: URI query parameters supported by the DELETE method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.2.2.3.3.1-2 and the response data structures and response codes specified in table 6.2.2.2.3.3.1-3.

Table 6.2.2.2.3.3.1-2: Data structures supported by the DELETE Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.2.2.2.3.3.1-3: Data structures supported by the DELETE Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	Successful case: The Individual Y1 RAI Subscription resource matching the subscriptionId is deleted.
NOTE 1: The mandatory HTTP error status codes for the DELETE method used by [7] also apply. See clause 5.2.5.				

6.2.2.2.3.3.2 HTTP PUT

This method shall support the URI query parameters specified in table 6.2.2.2.3.3.2-1.

Table 6.2.2.2.3.3.2-1: URI query parameters supported by the PUT method on this resource

Name	Data type	P	Cardinality	Description
n/a				

This method shall support the request data structures specified in table 6.2.2.2.3.3.2-2 and the response data structures and response codes specified in table 6.2.2.2.3.3.2-3.

Table 6.2.2.2.3.3.2-2: Data structures supported by the PUT Request Body on this resource

Data type	P	Cardinality	Description
RaiSubscription	M	1	Parameters to update an existing Individual Y1 RAI Subscription resource.

Table 6.2.2.2.3.3.2-3: Data structures supported by the PUT Response Body on this resource

Data type	P	Cardinality	Response codes	Description
RaiSubscription	M	1	200 OK	The Individual Y1 RAI Subscription resource is modified successfully and a representation of that resource is returned.
n/a			204 No Content	The Individual Y1 RAI Subscription resource is modified successfully.
ProblemDetails	O	0..1	500 Internal Server Error	(NOTE 2)
NOTE 1: The mandatory HTTP error status codes for the PUT method used by [7] also apply. See clause 5.2.5. NOTE 2: Failure cases are described in clause 6.2.2.6.				

6.2.2.2.3.4 Resource Custom Operations

No custom operations are defined in the present document.

6.2.2.3 Custom operations without associated resources

No custom operations are defined in the present document.

6.2.2.4 Notifications

6.2.2.4.1 General

The notifications are summarized in Table 6.2.2.4.1-1.

Table 6.2.2.4.1-1: Notifications overview

Notification	Callback URI	Mapped HTTP method	Description (service operation)
RAI Notification	{notificationTargetAddress}	POST	RAI_Notify operation.

6.2.2.4.2 RAI Notification

6.2.2.4.2.1 Description

The RAI Notification is used by the Near-RT RIC to report RAI to a Y1 consumer that has subscribed to such RAI associated with an Individual Y1 RAI Subscription resource, or to notify termination of the RAI subscription.

6.2.2.4.2.2 Operation Definition

Callback URI: **{notificationTargetAddress}**

The operation shall support the URI variables defined in table 6.2.2.4.2.2-1, the request data structures specified in table 6.2.2.4.2.2-2 and the response data structure and response codes specified in table 6.2.2.4.2.2-3.

Table 6.2.2.4.2.2-1: URI variables

Name	Data type	Definition
notificationTargetAddress	Uri	The Notification URI as assigned within the Individual Y1 RAI Subscription resource and described within the RAISubscription type.

Table 6.2.2.4.2.2-2: Data structures supported by the POST Request Body on this resource

Data type	P	Cardinality	Description
RaiNotification	M	1	Provides RAI or notifies termination of the RAI subscription.

Table 6.2.2.4.2.2-3: Data structures supported by the POST Response Body on this resource

Data type	P	Cardinality	Response codes	Description
n/a			204 No Content	The RAI Notification is acknowledged by the receiver.
NOTE: The mandatory HTTP error status codes for the DELETE method used by [7] also apply. See clause 5.2.5.				

6.2.2.5 Data model

6.2.2.5.1 General

This clause specifies the application data model.

Table 6.2.2.5.1-1 specifies the data types defined for the Y1_RAI_Subscription API in this specification.

Table 6.2.2.5.1-1: Y1_RAI_Subscription specific data types in this specification

Data type	Clause	Description	Applicability
NotificationCriteria	6.2.2.5.2.3	Contains the timing or the event that triggers RAI notifications.	
NotificationMethod	6.2.2.5.3.3	Represents the method used for RAI notifications.	
RaiNotification	6.2.2.5.2.4	Represents an individual occurrence of RAI notification.	
RaiReport	6.2.2.5.2.5	Represents a RAI report in a RAI notification.	
RaiSubscription	6.2.2.5.2.2	Represents an individual RAI subscription.	
TerminationCause	6.2.2.5.3.4	Indicates the cause from the Near-RT RIC to terminate a RAI subscription.	
ValidityPeriod	6.2.2.5.2.7	Indicate a validity period in UTC time.	
ValidityPeriodRelative	6.2.2.5.2.6	Indicate a validity period relative to a RAI notification.	

Table 6.2.2.5.1-2 specifies the data types re-used by the Y1_RAI_Subscription API from other specifications.

Table 6.2.2.5.1-2: Y1_RAI_Subscription re-used Data Types

Data type	Reference	Comments	Applicability
DateTime	3GPP TS 29.571 [8]	Identifies the actual date and time.	
FilterParameters	Y1TD [2]	Describes the filter(s) used for the RAI contents. This data type is defined per RAI type.	
NotificationTriggerEvent	Y1TD [2]	Contains the event that triggers RAI notifications. This data type is defined per RAI type.	
RAIContents	Y1TD [2]	Contains the actual RAI contents. This data type is defined per RAI type.	
RaiTypeId	Y1TD [2]	Indicates the type of RAI.	
RaiTypeVersion	Y1TD [2]	Indicates the version of the RAI type.	
TargetEntity	Y1TD [2]	Contains information used to identify a target entity with which a RAI report is associated. This data type is defined per RAI type.	

6.2.2.5.2 Structured data types

6.2.2.5.2.1 Introduction

This clause defines data structures to be used in resource representations.

6.2.2.5.2.2 Type RaiSubscription

Table 6.2.2.5.2.2-1: Definition of type RaiSubscription

Attribute name	Data type	P	Cardinality	Description	Applicability
raiType	RaiTypeId	M	1	The type of RAI.	
raiTypeVersion	RaiTypeVersion	M	1	Version of the RAI type.	
targetEntities	array(TargetEntity)	M	1..N	Information that identifies the entity/entities to which the requested RAI is related.	
expectedValidityPeriodsRelative	array(ValidityPeriodRelative)	O	1..N	Represents the expected validity period(s) for the RAI, which is relative to the occurrence of a RAI notification.	
filterParameters	FilterParameters	O	1	The conditions for filtering RAI contents to be included in a RAI notification.	
notificationCriteria	NotificationCriteria	M	1	Represents the timing or the events when the RAI notifications are triggered.	
notificationTargetAddress	Uri	C	0..1	Identifies the recipient of RAI notifications sent by the Near-RT RIC. This parameter shall be supplied by the Y1 consumer in the HTTP POST requests that create the subscriptions for RAI notifications, and in the HTTP PUT requests that update the subscriptions for RAI notifications.	

6.2.2.5.2.3 Type NotificationCriteria

Table 6.2.2.5.2.3-1: Definition of type NotificationCriteria

Attribute name	Data type	P	Cardinality	Description	Applicability
notificationMethod	NotificationMethod	M	1	Represents the notification method, e.g., periodic.	
notificationPeriod	integer	C	0..1	Indicates the periodicity of RAI notifications, in units of milliseconds. Shall be present when the "notificationMethod" attribute is set to "PERIODIC".	
notificationStartTime	DateTime	O	0..1	UTC time indicating the expected time of the first RAI notification. May be present when the "notificationMethod" attribute is set to "PERIODIC".	
notificationTriggerEvent	NotificationTriggerEvent	C	0..1	Represents the trigger event for RAI notifications. Shall be present when notificationMethod is set to "EVENT_TRIGGERED".	

6.2.2.5.2.4 Type RaiNotification

Table 6.2.2.5.2.4-1: Definition of type RaiNotification

Attribute name	Data type	P	Cardinality	Description	Applicability
subscriptionId	String	M	1	Identifies the RAI subscription with which the RAI notification is associated.	
terminationIndication	boolean	C	0..1	Indicates the termination of the RAI subscription by the Near-RT RIC. Shall be present and set to "TRUE" when the Near-RT RIC decides to terminate the subscription.	
terminationCause	TerminationCause	O	0..1	Indicates the reason why Near-RT RIC terminates the RAI subscription. May be present when the "terminationIndication" attribute is set to "TRUE".	
raiReports	array(RaiReport)	C	1..N	Represents the RAI reports. Shall be present when the "terminationIndication" attribute is absent or set to "FALSE".	

6.2.2.5.2.5 Type RaiReport

Table 6.2.2.5.2.5-1: Definition of type RaiReport

Attribute name	Data type	P	Cardinality	Description	Applicability
targetEntity	TargetEntity	M	1	The type of RAI.	
validityPeriod	ValidityPeriod	M	1	The validity period of the RAI report.	
raiContents	RaiContents	M	1	Contains the actual RAI contents.	
confidence	integer	O	0..1	Indicates the confidence of a prediction, in unit of percent. Minimum = 0. Maximum = 100.	
timeStampRaiGeneration	DateTime	O	0..1	Indicates the timestamp when the RAI report is generated by the Near-RT RIC.	

6.2.2.5.2.6 Type ValidityPeriodRelative

Table 6.2.2.5.2.6-1: Definition of type ValidityPeriodRelative

Attribute name	Data type	P	Cardinality	Description	Applicability
startTimeRelative	integer	M	1	Indicate the start time of the validity period in a relative manner, which is expressed as a time offset relative to the time of the RAI notification, in units of milliseconds. A negative value means a point of time before the RAI notification, and a positive value means a point of time after the RAI notification.	
durationMilliSecond	integer	O	0..1	Indicate the duration of validity period, in units of milliseconds. The absence of the attribute means a zero duration, i.e., the RAI is calculated only for the point of time given by the start time of the validity period.	

6.2.2.5.2.7 Type ValidityPeriod

Table 6.2.2.5.2.7-1: Definition of type ValidityPeriod

Attribute name	Data type	P	Cardinality	Description	Applicability
startTime	DateTime	M	1	Indicate the start time of the validity period in UTC time.	
durationMilliSecond	integer	O	0..1	Indicate the duration of validity period, in units of milliseconds. The absence of the attribute means a zero duration, i.e., the RAI is calculated only for the point of time given by the start time of the validity period.	

6.2.2.5.3 Simple data types and enumerations

6.2.2.5.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

6.2.2.5.3.2 Simple data types

The simple data types defined in table 6.2.2.5.3.2-1 shall be supported.

Table 6.2.2.5.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability
n/a			

6.2.2.5.3.3 Enumeration: NotificationMethod

Table 6.2.2.5.3.3-1: Enumeration NotificationMethod

Enumeration value	Description	Applicability
PERIODIC	The RAI notifications are sent with certain time interval.	
EVENT_TRIGGERED	The RAI notifications are triggered by certain event.	

6.2.2.5.3.4 Enumeration: TerminationCause

Table 6.2.2.5.3.4-1: Enumeration TerminationCause

Enumeration value	Description	Applicability
UNKNOWN	Indicates the RAI subscription is terminated due to unknown reasons.	
OVERLOAD	Indicates the RAI subscription is terminated due to overload in the Near-RT RIC.	

6.2.2.6 Error handling

6.2.2.6.1 General

The generic mechanism for error handling is described in clause 5.2.5. In addition, the requirements in the following clause 6.2.2.6.2 and clause 6.2.2.6.3 shall be applied.

6.2.2.6.2 Protocol Errors

In the present document, no additional protocol errors are defined for the Y1_RAI_Subscription API.

6.2.2.6.3 Application Errors

The application errors specific to the Y1_RAI_Subscription API are listed in table 6.2.2.6.2-1. The Near-RT RIC shall include in the HTTP status code a "ProblemDetails" data structure with the "cause" attribute indicating the application error as listed in table 6.2.2.6.2-1.

Table 6.2.2.6.2-1: Application errors

Application Error	HTTP status code	Description
DATA_UNAVAILABLE	500 Internal Server Error	Indicates the RAI subscription is rejected since necessary data to perform the analysis is unavailable.

6.3 Y1_RAI_Query service API

6.3.1 Introduction

The Y1_RAI_Query API is used to access the Y1_RAI_Query service.

The service operations and corresponding API definitions are organized in the following clauses for different solution sets.

6.3.2 HTTP REST with JSON solution set

6.3.2.1 Service operations

6.3.2.1.1 Introduction

Table 6.3.2.1.1-1 lists the operations supported by the Y1_RAI_Query service for the solution set "HTTP REST with JSON".

Table 6.3.2.2.1-1: Operations of the Y1_RAI_Subscription service

Service operation name	Description	Initiated by
RAI_Query	This service operation is used by a Y1 consumer to query specific RAI from Near-RT RIC.	Y1 consumer

6.3.2.1.2 RAI_Query service operation

6.3.2.1.2.1 General

The RAI_Query service operation is used by a Y1 consumer to query specific RAI from the Near-RT RIC.

6.3.2.1.2.2 Query RAI

Figure 6.3.2.1.2.2-1 shows a scenario where the Y1 consumer sends a request to the Near-RT RIC to query RAI. The operation is based on HTTP GET.

```
@startuml
participant "Y1 consumer" as cons
participant "Near-RT RIC" as prod
cons ->> prod : 1. GET .../analytics?query_parameters
prod ->> cons : 2. 200 OK
@enduml
```


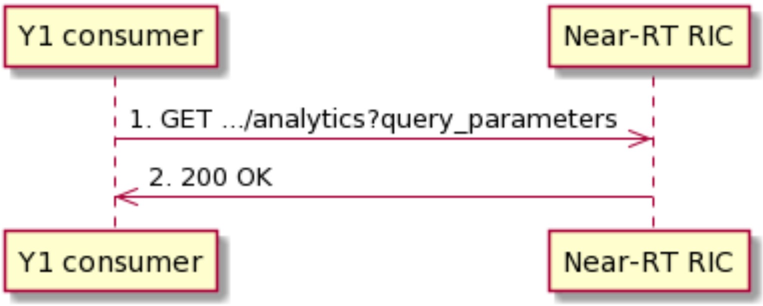

Figure 6.3.2.1.2.2-1: Y1 consumer query a RAI

The Y1 consumer shall send an HTTP GET request to the Near-RT RIC on the resource URI "{apiRoot}/Y1_RAI_Query/<apiMajorVersion>/analytics" representing the "Y1 RAI" resource, to request RAI.

NOTE: The path segment " Y1_RAI_Query " does not follow the related naming convention specified in clause 5.2.4.1 of [7]. The path segment is however kept as currently defined in this specification for backward compatibility considerations.

The following information shall be provided:

- the type of RAI as the "raiType" attribute;
- the version of the RAI type as the "raiTypeVersion" attribute;
- the target entity/entities of the RAI in the "targetEntities" attribute.

The following information may be provided:

- the expected validity period(s) for the RAI in the "expectedValidityPeriods" attribute; an expected validity period before the occurrence of a RAI query refers to statistics for a past time, and an expected validity period after the occurrence of a RAI query refers to a prediction for a future time.
- the filter parameters to filter RAI contents in the RAI report as the "filterParameters" attribute;

Upon the reception of the HTTP GET request, the Near-RT RIC shall analyse RAI according to the request.

If the HTTP request message from the Y1 consumer is accepted, the Near-RT RIC shall respond with "200 OK" status code with the message body containing the RAI with parameters as relevant for the requesting Y1 consumer. The "raiQueryResult" data structure in the response body shall include:

- the RAI reports in the "raiReports" attribute that, for each RAI report, the "RaiReport" data type shall include:
    - the target entity of the RAI report as the "targetEntity" attribute;
    - the validity period of the RAI report as the "validityPeriod" attribute;
    - the actual RAI contents as the "raiContents" attribute;

  and may include:

    - the confidence of the RAI report, if the RAI report is a prediction;
    - the timestamp when Near-RT RIC generates the RAI report as the "timeStampRaiGeneration" attribute.

If an error occurs when processing the HTTP POST request, the Near-RT RIC shall send an HTTP error response as specified in clause 6.2.2.6. In particular,

- If the necessary data to perform the analysis are unavailable, the Near-RT RIC shall reject the request with a "500 Internal Server Error" status code and a "ProblemDetails" data structure including the "cause" attribute set to "DATA_UNAVAILABLE".

6.3.2.2 Resources

6.3.2.2.1 Resource URI structure

The resource URI shall have the structure as defined in clause 5.2.3.1 with the following clarifications:

- The <apiName> shall be "Y1_RAI_Query".
- The <apiMajorVersion> shall be "v1".
- The <apiSpecificResourceUriPart> shall be set as illustrated in figure 6.3.2.2.1-1.

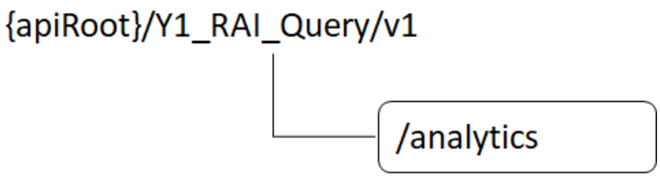

Figure 6.3.2.2.1-1: Resource URI structure of the Y1_RAI_Query API

Table 6.3.2.2.1-1 provides an overview of the resources and applicable HTTP methods for the Y1_RAI_Query API.

Table 6.3.2.2.1-1: Resources and methods overview

Resource name	API specific resource URI part	HTTP method or custom operation	Description
Y1 RAI	/analytics	GET	Retrieve the RAI

6.3.2.2.2 Resource: Y1 RAI

6.3.2.2.2.1 Description

The Y1 RAI resource represents the RAI to the RAI_Query Service at a given Near-RT RIC.

6.3.2.2.2.2 Resource Definition

Resource URI: **{apiRoot}/Y1_RAI_Query/v1/analytics**

This resource shall support the resource URI variables defined in table 6.3.2.2.2.2-1.

Table 6.3.2.2.2.2-1: Resource URI variables for this resource

Name	Data type	Definition
apiRoot	string	See clause 5.2.3.1

6.3.2.2.2.3 Resource Standard Methods

6.3.2.2.2.3.1 HTTP GET

This method shall support the URI query parameters specified in table 6.3.2.2.2.3.1-1.

Table 6.3.2.2.2.3.1-1: URI query parameters supported by the GET method on this resource

Name	Data type	P	Cardinality	Description
raiType	RaiTypeId	M	1	Indicates the type of RAI.
raiTypeVersion	RaiTypeVersion	M	1	Indicates the version of the RAI type.
targetEntities	array(TargetEntity)	M	1..N	Information that identifies the entity/entities to which the requested RAI is related.
expectedValidityPeriods	array(ValidityPeriod)	O	1..N	Represents the expected validity period(s) for the RAI.
filterParameters	FilterParameters	O	1	Describes the filter(s) used for the RAI contents.
NOTE: The query parameters "raiType", "raiTypeVersion", "targetEntities", "expectedValidityPeriods" and "filterParameters" do not follow the related naming convention specified in clause 5.2.4.1 of [7]. These query parameters are however kept as currently defined in this specification for backward compatibility considerations.				

This method shall support the request data structures specified in table 6.3.2.2.2.3.1-2 and the response data structures and response codes specified in table 6.3.2.2.2.3.1-3.

Table 6.3.2.2.2.3.1-2: Data structures supported by the GET Request Body on this resource

Data type	P	Cardinality	Description
n/a			

Table 6.3.2.2.2.3.1-3: Data structures supported by the GET Response Body on this resource

Data type	P	Cardinality	Response codes	Description
RaiQueryResult	M	1	200 OK	Containing the RAI with parameters as relevant for the requesting Y1 consumer
ProblemDetails	O	0..1	500 Internal Server Error	(NOTE 2)
NOTE 1: The mandatory HTTP error status codes for the GET method used by [7] also apply. See clause 5.2.5. NOTE 2: Failure cases are described in clause 6.3.2.6.				

6.3.2.3 Custom operations without associated resources

No custom operations are defined in the present document.

6.3.2.4 Notifications

No notifications are defined in the present document.

6.3.2.5 Data model

6.3.2.5.1 General

Table 6.3.2.5.1-1 specifies the data types defined for the Y1_RAI_Query API in this specification.

Table 6.3.2.5.1-1: Y1_RAI_Query specific data types in this specification

Data type	Clause	Description	Applicability
RaiQueryResult	6.3.2.5.2.2	Contains the RAI with parameters indicated in the request.	

Table 6.3.2.5.1-2 specifies the data types re-used by the Y1_RAI_Query API from other specifications.

Table 6.3.2.5.1-2: Y1_RAI_Query re-used Data Types

Data type	Reference	Comments	Applicability
DateTime	3GPP TS 29.571 [8]	Identifies the actual date and time.	
FilterParameters	Y1TD [2]	Describes the filter(s) used for the RAI contents. This data type is defined per RAI type.	
RAIContents	Y1TD [2]	Contains the actual RAI contents. This data type is defined per RAI type.	
RaiReport	6.2.2.5.2.5	Represents a RAI report in a RAI query result.	
RaiTypeId	Y1TD [2]	Indicates the type of RAI.	
RaiTypeVersion	Y1TD [2]	Indicates the version of the RAI type.	
TargetEntity	Y1TD [2]	Contains information used to identify a target entity with which a RAI report is associated. This data type is defined per RAI type.	
ValidityPeriod	6.2.2.5.2.7	Indicate a validity period in UTC time.	

6.3.2.5.2 Structured data types

6.3.2.5.2.1 Introduction

This clause defines data structures to be used in resource representations.

6.3.2.5.2.2 Type RaiQueryResult

Table 6.3.2.5.2.2-1: Definition of type RaiResultSet

Attribute name	Data type	P	Cardinality	Description	Applicability
raiReports	array(RaiReport)	M	1..N	Represents the RAI reports.	

6.3.2.5.3 Simple data types and enumerations

6.3.2.5.3.1 Introduction

This clause defines simple data types and enumerations that can be referenced from data structures defined in the previous clauses.

6.3.2.5.3.2 Simple data types

The simple data types defined in table 6.3.2.5.3.2-1 shall be supported.

Table 6.3.2.5.3.2-1: Simple data types

Type Name	Type Definition	Description	Applicability
n/a			

6.3.2.6 Error handling

6.3.2.6.1 General

The generic mechanism for error handling is described in clause 5.2.5. In addition, the requirements in the following clauses shall apply.

6.3.2.6.2 Protocol Errors

In the present document, no additional protocol errors are defined for the Y1_RAI_Query API.

6.3.2.6.3 Application Errors

The application errors specific to the Y1_RAI_Query API are listed in table 6.3.2.6.3-1. The Near-RT RIC shall include in the HTTP status code a "ProblemDetails" data structure with the "cause" attribute indicating the application error as listed in table 6.2.2.6.3-1.

Table 6.3.2.6.3-1: Application errors

Application Error	HTTP status code	Description
DATA_UNAVAILABLE	500 Internal Server Error	Indicates the RAI subscription is rejected since necessary data to perform the analysis is unavailable.

Annex A (normative): OpenAPI specification

A.1 General

This Annex specifies the formal definition of the Y1 service API(s). It consists of OpenAPI documents in YAML format that are based on the OpenAPI Specification [2].

A.2 Y1_RAI_Subscription API

```
openapi: 3.1.0

info:
  title: Y1_RAI_Subscription API
  description: |
    Y1_RAI_Subscription API as part of Y1 interface.
    © 2023, O-RAN ALLIANCE
    All rights reserved.
  version: "1.0.0"

externalDocs:
  description: O-RAN.WG3.Y1AP-v01.00
  url: 'https://www.o-ran.org/specifications'

servers:
  - url: '{apiRoot}/Y1_RAI_Subscription/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 5.2.3 of O-RAN.WG3.Y1AP-v01.00

paths:
  /subscriptions:
    post:
      description: Creates a new Individual Y1 RAI Subscription.
      operationId: CreateY1RAISubscription
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/RaiSubscription'
      responses:
        '201':
          description: >
            The Individual Y1 RAI Subscription resource is created successfully and a
            representation of that resource is returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/RaiSubscription'
          headers:
            Location:
              description: >
                Contains the URI of the newly created resource, according to the structure
                {apiRoot}/Y1_RAI_Subscription/v1/subscriptions/{subscriptionId}
              required: true
              schema:
                type: string
        '400':
          $ref: '#/components/responses/400'
        '401':
          $ref: '#/components/responses/401'
```

```yaml
        '403':
          $ref: '#/components/responses/403'
        '404':
          $ref: '#/components/responses/404'
        '411':
          $ref: '#/components/responses/411'
        '413':
          $ref: '#/components/responses/413'
        '415':
          $ref: '#/components/responses/415'
        '429':
          $ref: '#/components/responses/429'
        '500':
          $ref: '#/components/responses/500'
        '503':
          $ref: '#/components/responses/503'
        default:
          $ref: '#/components/responses/default'
      callbacks:
        RAINotification:
          '{$request.body#/notificationTargetAddress}':
            post:
              requestBody:  # contents of the callback message
                required: true
                content:
                  application/json:
                    schema:
                      $ref: '#/components/schemas/RaiNotification'
              responses:
                '204':
                  description: No Content (successful notification)
                '307':
                  $ref: '#/components/responses/307'
                '308':
                  $ref: '#/components/responses/308'
                '400':
                  $ref: '#/components/responses/400'
                '401':
                  $ref: '#/components/responses/401'
                '403':
                  $ref: '#/components/responses/403'
                '404':
                  $ref: '#/components/responses/404'
                '411':
                  $ref: '#/components/responses/411'
                '413':
                  $ref: '#/components/responses/413'
                '415':
                  $ref: '#/components/responses/415'
                '429':
                  $ref: '#/components/responses/429'
                '500':
                  $ref: '#/components/responses/500'
                '503':
                  $ref: '#/components/responses/503'
                default:
                  $ref: '#/components/responses/default'

  /subscriptions/{subscriptionId}:
    delete:
      description: Deletes an Individual Y1 RAI Subscription.
      operationId: DeleteY1RAISubscription
      parameters:
        - name: subscriptionId
          in: path
          description: Identifier of the Individual Y1 RAI Subscription
          required: true
          schema:
            type: string
      responses:
```



```
        '204':
          description: >
            The Individual Y1 RAI Subscription resource matching the subscriptionId is deleted.
        '307':
          $ref: '#/components/responses/307'
        '308':
          $ref: '#/components/responses/308'
        '400':
          $ref: '#/components/responses/400'
        '401':
          $ref: '#/components/responses/401'
        '403':
          $ref: '#/components/responses/403'
        '404':
          $ref: '#/components/responses/404'
        '429':
          $ref: '#/components/responses/429'
        '500':
          $ref: '#/components/responses/500'
        '503':
          $ref: '#/components/responses/503'
        default:
          $ref: '#/components/responses/default'
    put:
      description: Updates an Individual Y1 RAI Subscription
      operationId: UpdateY1RAISubscription
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/RaiSubscription'
      parameters:
        - name: subscriptionId
          in: path
          description: Identifier of the Individual Y1 RAI Subscription
          required: true
          schema:
            type: string
      responses:
        '200':
          description: >
            The Individual Y1 RAI Subscription resource is updated successfully and a
            representation of that resource is returned.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/RaiSubscription'
        '204':
          description: The Individual Y1 RAI Subscription resource is updated successfully.
        '307':
          $ref: '#/components/responses/307'
        '308':
          $ref: '#/components/responses/308'
        '400':
          $ref: '#/components/responses/400'
        '401':
          $ref: '#/components/responses/401'
        '403':
          $ref: '#/components/responses/403'
        '404':
          $ref: '#/components/responses/404'
        '411':
          $ref: '#/components/responses/411'
        '413':
          $ref: '#/components/responses/413'
        '415':
          $ref: '#/components/responses/415'
        '429':
          $ref: '#/components/responses/429'
```

```
        '500':
          $ref: '#/components/responses/500'
        '503':
          $ref: '#/components/responses/503'
        default:
          $ref: '#/components/responses/default'

components:
  schemas:
    RaiSubscription:
      type: object
      description: Represents an Individual RAI Subscription resource.
      properties:
        raiType:
          $ref: '#/components/schemas/RaiTypeId'
        raiTypeVersion:
          $ref: '#/components/schemas/RaiTypeVersion'
        targetEntities:
          type: array
          items:
            $ref: '#/components/schemas/TargetEntity'
          minItems: 1
        expectedValidityPeriodsRelative:
          type: array
          items:
            $ref: '#/components/schemas/ValidityPeriodRelative'
        filterParameters:
          $ref: '#/components/schemas/FilterParameters'
        notificationCriteria:
          $ref: '#/components/schemas/NotificationCriteria'
        notificationTargetAddress:
          $ref: '#/components/schemas/Uri'
      required:
        - raiType
        - raiTypeVersion
        - notificationCriteria
        - targetEntities

    NotificationCriteria:
      type: object
      description: Contains the timing or event that triggers RAI notifications.
      properties:
        notificationMethod:
          $ref: '#/components/schemas/NotificationMethod'
        notificationPeriod:
          type: integer
          description: Indicates the interval of periodic RAI notifications in units of milliseconds.
        notificationStartTime:
          $ref: '#/components/schemas/DateTime'
          description: Indicates the expected time of the first notification for the periodic RAI
notifications.
        notificationTriggerEvent:
          $ref: '#/components/schemas/NotificationTriggerEvent'
      required:
        - notificationMethod

    RaiNotification:
      type: object
      description: Represents a RAI notification.
      properties:
        subscriptionId:
          type: string
          description: >
            Identifier of the subscription resource to which the RAI notification
            is related.
        terminationIndication:
          type: boolean
          description: When true, indicates the termination of the RAI subscription by the Near-RT RIC.
        terminationCause:
          $ref: '#/components/schemas/TerminationCause'
```



```
        raiReports:
          type: array
          items:
            $ref: '#/components/schemas/RaiReport'
          minItems: 1
      required:
        - subscriptionId

    RaiTypeId:
      type: string
      description: Indicates the type of RAI. It is defined in Y1TD.

    RaiTypeVersion:
      type: string
      description: Indicates the version of the RAI type. It is defined in Y1TD.

    RaiContents:
      type: object
      description: Contains the actual RAI contents. It is defined in Y1TD per RAI type.

    RaiReport:
      type: object
      description: Represents a RAI report in a RAI notification.
      properties:
        targetEntity:
          $ref: '#/components/schemas/TargetEntity'
        validityPeriod:
          $ref: '#/components/schemas/ValidityPeriod'
        raiContents:
          $ref: '#/components/schemas/RaiContents'
        confidence:
          type: integer
          minimum: 0
          maximum: 100
          description: Indicates the confidence of a prediction, in unit of percent.
        timeStampRaiGeneration:
          $ref: '#/components/schemas/DateTime'
          description: The date and time when the RAI report is generated.
      required:
        - targetEntity
        - validityPeriod
        - raiContents

    TargetEntity:
      type: object
      description: Contains information used to identify a target entity with which a RAI report is
associated. It is defined in Y1TD per RAI type.

    TerminationCause:
      type: string
      enum:
        - UNKNOWN
        - OVERLOAD
      description: Indicates the cause from the Near-RT RIC to terminate a RAI subscription.

    ValidityPeriodRelative:
      type: object
      description: Indicate a validity period relative to a RAI notification.
      properties:
        startTimeRelative:
          type: integer
          description: Indicate the start time of the validity period in a relative manner, which is
expressed as a time offset relative to the time of the RAI notification, in units of milliseconds
        durationMilliSecond:
          type: integer
          description: Indicate the duration of validity period, in units of milliseconds.
      required:
        - startTimeRelative

    ValidityPeriod:
```

```
      type: object
      description: Indicate a validity period in UTC time.
      properties:
        startTime:
          $ref: '#/components/schemas/DateTime'
        durationMilliSecond:
          type: integer
          description: Indicate the duration of validity period, in units of milliseconds.
      required:
        - startTime

    FilterParameters:
      type: object
      description: Describes the filter(s) used for the RAI contents. It is defined in Y1TD per RAI type.

    NotificationMethod:
      type: string
      enum:
        - PERIODIC
        - EVENT_TRIGGERED
      description: Represents the method used for RAI notifications.

    NotificationTriggerEvent:
      type: object
      description: Contains the event that triggers RAI notifications. It is defined in Y1TD per RAI type.

    DateTime:
      type: string
      format: date-time
      description: date and time as defined by RFC3339.

    Uri:
      type: string
      description: string providing an URI formatted according to IETF RFC 3986.

    ProblemDetails:
      type: object
      properties:
        type:
          $ref: '#/components/schemas/Uri'
        title:
          type: string
          description: A short, human-readable summary of the problem type. It should not change from
occurrence to occurrence of the problem.
        status:
          type: integer
          description: The HTTP status code for this occurrence of the problem.
        detail:
          type: string
          description: A human-readable explanation specific to this occurrence of the problem.
        instance:
          $ref: '#/components/schemas/Uri'
        cause:
          type: string
          description: A machine-readable application error cause specific to this occurrence of the
problem. This IE should be present and provide application-related error information, if available.
        invalidParams:
          type: array
          items:
            $ref: '#/components/schemas/InvalidParam'
          minItems: 1
          description: Description of invalid parameters, for a request rejected due to invalid
parameters.

    InvalidParam:
      type: object
      properties:
        param:
          type: string
          description: Attribute's name encoded as a JSON Pointer, or header's name.
```

```yaml
        reason:
          type: string
          description: A human-readable reason, e.g. "must be a positive integer".
      required:
        - param
#
# HTTP Responses
#
responses:
  '307':
    description: Temporary Redirect
    headers:
      Location:
        description: 'An alternative URI of the resource.'
        required: true
        schema:
          type: string
  '308':
    description: Permanent Redirect
    headers:
      Location:
        description: 'An alternative URI of the resource.'
        required: true
        schema:
          type: string
  '400':
    description: Bad request
    content:
      application/problem+json:
        schema:
          $ref: '#/components/schemas/ProblemDetails'
  '401':
    description: Unauthorized
    content:
      application/problem+json:
        schema:
          $ref: '#/components/schemas/ProblemDetails'
  '403':
    description: Forbidden
    content:
      application/problem+json:
        schema:
          $ref: '#/components/schemas/ProblemDetails'
  '404':
    description: Not Found
    content:
      application/problem+json:
        schema:
          $ref: '#/components/schemas/ProblemDetails'
  '406':
    description: Not Acceptable
    content:
      application/problem+json:
        schema:
          $ref: '#/components/schemas/ProblemDetails'
  '409':
    description: Conflict
    content:
      application/problem+json:
        schema:
          $ref: '#/components/schemas/ProblemDetails'
  '411':
    description: Length Required
    content:
      application/problem+json:
        schema:
          $ref: '#/components/schemas/ProblemDetails'
  '412':
    description: Precondition Failed
    content:
```

```
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '413':
      description: Payload Too Large
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '414':
      description: URI Too Long
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '415':
      description: Unsupported Media Type
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '429':
      description: Too Many Requests
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '500':
      description: Internal Server Error
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '503':
      description: Service Unavailable
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    default:
      description: Generic Error
```

A.3 Y1_RAI_Query API

```
openapi: 3.1.0

info:
  title: Y1_RAI_Query API
  description: |
    Y1_RAI_Query API as part of Y1 interface.
    © 2023, O-RAN ALLIANCE
    All rights reserved.
  version: "1.0.0"

externalDocs:
  description: O-RAN.WG3.Y1AP-v01.00
  url: 'https://www.o-ran.org/specifications'

servers:
  - url: '{apiRoot}/Y1_RAI_Query/v1'
    variables:
      apiRoot:
        default: https://example.com
        description: apiRoot as defined in clause 5.2.3 of O-RAN.WG3.Y1AP-v01.00

paths:
  /analytics:
    get:
      description: Retrieve RAI.
```

```
      operationId: GetY1RAI
      parameters:
        - name: raiType
          in: query
          description: Identify the type of RAI.
          required: true
          schema:
            $ref: '#/components/schemas/RaiTypeId'
        - name: raiTypeVersion
          in: query
          description: Identify the version of the RAI type.
          required: true
          schema:
            $ref: '#/components/schemas/RaiTypeVersion'
        - name: targetEntities
          in: query
          description: Identify the entity/entities to which the requested RAI is related.
          required: true
          schema:
            type: array
            items:
              $ref: '#/components/schemas/TargetEntity'
            minItems: 1
        - name: expectedValidityPeriods
          in: query
          description: Represents the expected validity period(s) for the RAI.
          required: false
          schema:
            type: array
            items:
              $ref: '#/components/schemas/ValidityPeriod'
        - name: filterParameters
          in: query
          description: Identify the filter(s) used for the RAI contents.
          required: false
          schema:
            $ref: '#/components/schemas/FilterParameters'
      responses:
        '200':
          description: >
            Containing the RAI query result with parameters as relevant for the requesting Y1 consumer.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/RaiQueryResult'
        '400':
          $ref: '#/components/responses/400'
        '401':
          $ref: '#/components/responses/401'
        '403':
          $ref: '#/components/responses/403'
        '404':
          description: Indicates that the Y1 RAI resource does not exist.
          content:
            application/problem+json:
              schema:
                $ref: '#/components/schemas/ProblemDetails'
        '406':
          $ref: '#/components/responses/406'
        '414':
          $ref: '#/components/responses/414'
        '429':
          $ref: '#/components/responses/429'
        '500':
          $ref: '#/components/responses/500'
        '503':
          $ref: '#/components/responses/503'
        default:
          $ref: '#/components/responses/default'
```



```
components:
  schemas:
    RaiQueryResult:
      type: object
      description: Represents the RAI with parameters as relevant for the requesting Y1 consumer.
      properties:
        raiReports:
          type: array
          items:
            $ref: '#/components/schemas/RaiReport'
          minItems: 1
      required:
        - raiReports

    RaiTypeId:
      type: string
      description: Indicates the type of RAI. It is defined in Y1TD.

    RaiTypeVersion:
      type: string
      description: Indicates the version of the RAI type. It is defined in Y1TD.

    RaiContents:
      type: object
      description: Contains the actual RAI contents. It is defined in Y1TD per RAI type.

    RaiReport:
      type: object
      description: Represents a RAI report in a RAI notification.
      properties:
        targetEntity:
          $ref: '#/components/schemas/TargetEntity'
        validityPeriod:
          $ref: '#/components/schemas/ValidityPeriod'
        raiContents:
          $ref: '#/components/schemas/RaiContents'
        confidence:
          type: integer
          minimum: 0
          maximum: 100
          description: Indicates the confidence of a prediction, in unit of percent.
        timeStampRaiGeneration:
          $ref: '#/components/schemas/DateTime'
          description: The date and time when the RAI report is generated.
      required:
        - targetEntity
        - validityPeriod
        - raiContents

    FilterParameters:
      type: object
      description: Describes the filter(s) used for the RAI contents. It is defined in Y1TD per RAI type.

    TargetEntity:
      type: object
      description: Contains information used to identify a target entity with which a RAI report is
associated. It is defined in Y1TD per RAI type.

    ValidityPeriod:
      type: object
      description: Indicate a validity period in UTC time.
      properties:
        startTime:
          $ref: '#/components/schemas/DateTime'
        durationMilliSecond:
          type: integer
          description: Indicate the duration of validity period, in units of milliseconds.
      required:
        - startTime
```



```
    DateTime:
      type: string
      format: date-time
      description: date and time as defined by RFC3339.

    Uri:
      type: string
      description: string providing an URI formatted according to IETF RFC 3986.

    ProblemDetails:
      type: object
      properties:
        type:
          $ref: '#/components/schemas/Uri'
        title:
          type: string
          description: A short, human-readable summary of the problem type. It should not change from
occurrence to occurrence of the problem.
        status:
          type: integer
          description: The HTTP status code for this occurrence of the problem.
        detail:
          type: string
          description: A human-readable explanation specific to this occurrence of the problem.
        instance:
          $ref: '#/components/schemas/Uri'
        cause:
          type: string
          description: A machine-readable application error cause specific to this occurrence of the
problem. This IE should be present and provide application-related error information, if available.
        invalidParams:
          type: array
          items:
            $ref: '#/components/schemas/InvalidParam'
          minItems: 1
          description: Description of invalid parameters, for a request rejected due to invalid
parameters.

    InvalidParam:
      type: object
      properties:
        param:
          type: string
          description: Attribute's name encoded as a JSON Pointer, or header's name.
        reason:
          type: string
          description: A human-readable reason, e.g. "must be a positive integer".
      required:
        - param
  #
  # HTTP Responses
  #
  responses:
    '307':
      description: Temporary Redirect
      headers:
        Location:
          description: 'An alternative URI of the resource.'
          required: true
          schema:
            type: string
    '308':
      description: Permanent Redirect
      headers:
        Location:
          description: 'An alternative URI of the resource.'
          required: true
          schema:
            type: string
```



```
    '400':
      description: Bad request
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '401':
      description: Unauthorized
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '403':
      description: Forbidden
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '404':
      description: Not Found
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '406':
      description: Not Acceptable
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '409':
      description: Conflict
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '411':
      description: Length Required
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '412':
      description: Precondition Failed
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '413':
      description: Payload Too Large
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '414':
      description: URI Too Long
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '415':
      description: Unsupported Media Type
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '429':
      description: Too Many Requests
      content:
        application/problem+json:
```

```
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '500':
      description: Internal Server Error
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    '503':
      description: Service Unavailable
      content:
        application/problem+json:
          schema:
            $ref: '#/components/schemas/ProblemDetails'
    default:
      description: Generic Error
```


Annex (informative): Change History

Date	Revision	Description
2023.05.05	00.00.01	Create the skeleton.
2023.10.23	00.00.02	Implemented CMCC-2023.07.25-WG3-CR-0002-Y1AP-General contents for clause 4 and 5-v02.docx CMCC-2023.09.04-WG3-CR-0003-Y1AP-Y1_RAI_Subscription API-v02.docx CMCC-2023.10.07-WG3-CR-0004-Y1AP-Y1_RAI_Query API-v01.docx and editorial changes.
2023.10.31	00.00.03	Implemented CUC-2023.10.27-WG3-CR-0001-Correction-Of-RAI-Notify-v01.docx
2023.11.17	00.00.04	Editorial changes according to the review comments during WG3 poll.
2023.11.18	01.00	Version incremented for publication.
2024.07.08	01.00.01	Implemented CMCC-2024.05.24-WG3-CR-0005-Y1AP-Editorial corrections for ODR alignment-v01.docx and editorial changes.
2024.07.22	01.00.02	Editorial changes according to the review comments during WG3 poll.
2024.07.26	01.01	Version incremented for TSC.